Aim:

Image processing and recognition for color and shape detection (recognizing simple objects based on features)

Objective:

To understand basic concepts of **image processing, object recognition** and to detect objects based on **color features (HSV space)** shapes (triangle, square, circle) using **contour analysis**

Software Requirements

- MATLAB / Python
- Libraries:
  - Open CV
  - NumPy
  - Matplotlib

Theory

Image Processing

Image processing involves manipulating digital images to extract useful information such as edges, shapes, and colors.

Color Detection (HSV Model)

Instead of RGB, we use HSV (Hue, Saturation, Value):

- **Hue (H)** → represents color (e.g., red, yellow)
- **Saturation (S)** → intensity of color
- **Value (V)** → brightness

HSV is preferred because it separates color from lighting, making detection easier.

Shape Detection

Shapes are detected using **contours**:

- A **contour** is the boundary of an object
- Using approx.PolyDP, we approximate shape:
  - 3 sides → Triangle
  - 4 sides → Square
  - 6 sides → Circle

Feature-Based Recognition

Objects are recognized using:

- **Color features** (HSV range)
- **Geometric features** (number of edges)

Algorithm

1. Install and import required libraries
2. Load the image
3. Convert image to HSV
4. Define color range
5. Create mask
6. Detect contours
7. Identify shapes
8. Display output

Conclusion